

Mathematics A

Unit 2 • Chapter OL-T27 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 25 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T27 • Expertise Q16+ • 5 items • 25 marks

Chapter	Items	Marks	Starts
01. OL-T27 Simultaneous Equations	5	25	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Simultaneous Equations

Unit 2 • Expertise • 25 marks • 5 item(s)

June 2024 | 4MA1-2H Q22 | 5 marks

Line with mixed quadratic relation

November 2024 | 4MA1-2H Q22 | 5 marks

Line with circle or sum-of-squares curve

January 2020 | 4MA1-2H Q22 | 5 marks

Multiple solution-pair validation

May-Nov 2020 | 2H Q16 | 5 marks

Draw the missing line for an intersection

January 2023 | 2H Q20 | 5 marks

Draw the missing line for an intersection

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q22

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

WORKING SPACE

4MA1-2H Q22

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

Line-curve intersections

5 marks

Substitute

$$y = 5 - x : \quad 2x^2 + 3(5 - x)^2 = 210.$$

Solve

$$x^2 - 6x - 27 = 0 \Rightarrow x = 9, -3.$$

Coordinates

$$y = -4, 8.$$

Final answer

$$(9, -4), (-3, 8)$$

4MA1-2H Q22 M01

5 marks

22 Solve the simultaneous equations

$$x^2 + y^2 + y = 3$$

$$x + 2 = y$$

Show clear algebraic working.

WORKING SPACE

Solve simultaneous equations

5 marks

M01 – Solve simultaneous equations

Substitute $y=x+2$

$$x^2 + y^2 + y = 3 \Rightarrow x^2 + (x+2)^2 + (x+2) = 3.$$

Form the quadratic

$$2x^2 + 5x + 3 = 0 = (2x+3)(x+1).$$

Find x-values

$$x = -3/2 \text{ or } x = -1.$$

Find y-values

$$y = x + 2 \Rightarrow y = 1/2 \text{ or } y = 1.$$

Pair the solutions

$$(-3/2, 1/2) \text{ and } (-1, 1).$$

Final answer

$$\left(-\frac{3}{2}, \frac{1}{2}\right) \text{ and } (-1, 1)$$

- 22** The line with equation $y = x + 2$ intersects the curve with equation $x^2 + y^2 - 2y = 24$ at the points A and B .

Find the coordinates of A and B .

Show clear algebraic working.

(..... ,)

(..... ,)

Worked solution

January 2020 | Question 22 | 5 marks

Family: Solve a linear-quadratic system • Variant: Multiple solution-pair validation

Method / working

1. Substitute $y=x+2$ into $x^2+y^2-2y=24$.
2. Obtain $x^2+x-12=0$.
3. Factor: $(x+4)(x-3)=0$.
4. $x=-4$ gives $y=-2$; $x=3$ gives $y=5$.
5. State both points.

Final answer: $(-4,-2)$ and $(3,5)$

16 Solve the simultaneous equations

$$\begin{aligned}3xy - y^2 &= 8 \\ x - 2y &= 1\end{aligned}$$

Show clear algebraic working.

(Total for Question 16 is 5 marks)

Answer reference | May-Nov 2020 | Question 16

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

16	$3y(2y+1) - y^2 = 8$ or $x = \frac{8+y^2}{3y} \rightarrow \frac{8+y^2}{3y} - 2y = 1$ or $-3xy - y^2 = 8$ $3xy - 3y \times 2y = 3y \times 1$ oe	$3x\left(\frac{x-1}{2}\right) - \left(\frac{x-1}{2}\right)^2 = 8$ oe			M1 correct first step eg substitution by eg $x = 1 + 2y$ or $y = \frac{x-1}{2}$ to get an equation in a single variable or writing 2 nd equation with x the subject and substituting into 1 st or multiplying 2 nd equation by $3y$ and subtracting from 1 st oe
	eg $5y^2 + 3y - 8 (= 0)$	eg $5x^2 - 4x - 33 (= 0)$			A1 for a correct simplified quadratic
	$(5y+8)(y-1) (= 0)$ or $\frac{-3 \pm \sqrt{3^2 - 4 \times 5 \times (-8)}}{2 \times 5}$	$(5x+11)(x-3) (= 0)$ or $\frac{4 \pm \sqrt{(-4)^2 - 4 \times 5 \times (-33)}}{2 \times 5}$			M1ft dep on M1 for solving their 3 term quadratic equation using any correct method (allow one sign error and some simplification – allow as far as $\frac{-3 \pm \sqrt{9+160}}{10}$) or if factorising, allow brackets which expanded give 2 out of 3 terms correct)
	$y = -\frac{8}{5}$ and $y = 1$ (both)	$x = -\frac{11}{5}$ and $x = 3$ (both)			A1 dep on first M1
		$x = -\frac{11}{5}, y = -\frac{8}{5}$ $x = 3, y = 1$	5		A1 oe dep on first M1 Must be paired correctly
					Total 5 marks

Worked solution

May-Nov 2020 | Question 16 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.

20 Solve the simultaneous equations

$$\begin{aligned}y &= 7 - 2x \\ x^2 + y^2 &= 34\end{aligned}$$

Show clear algebraic working.

(Total for Question 20 is 5 marks)

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

20	$x^2 + (7 - 2x)^2 = 34$	$\left(\frac{7-y}{2}\right)^2 + y^2 = 34$	5	M1	substitution of linear equation into quadratic
	$5x^2 - 28x + 15 [= 0]$ oe	$5y^2 - 14y - 87 [= 0]$ oe		M1	dep on previous M1 for multiplying out and collecting terms, forming a three term quadratic in any form of $ax^2 + bx + c (= 0)$ where at least 2 coefficients (a or b or c) are correct and all are non-zero
	$(5x - 3)(x - 5) [= 0]$ or $\frac{-(-28) \pm \sqrt{(-28)^2 - 4 \times 5 \times 15}}{2 \times 5}$ or $5\left(x - \frac{28}{10}\right)^2 - \frac{784}{100} + 15 = 0$ oe or $x = 0.6$ and $x = 5$ (allow incorrect labels for x/y)	$(5y - 29)(y + 3) [= 0]$ or $\frac{-(-14) \pm \sqrt{(-14)^2 - 4 \times 5 \times (-87)}}{2 \times 5}$ or $5\left(y - \frac{14}{10}\right)^2 - \frac{196}{100} - 87 = 0$ oe or $y = 5.8$ and $y = -3$ (allow incorrect labels for x/y)		M1ft	dep on M1 for solving <i>their</i> 3 term quadratic equation using any correct method (if factorising, allow brackets which expanded give 2 out of 3 terms correct) (if using formula allow one sign error and some simplification – allow as far as $\frac{28 \pm \sqrt{784 - 300}}{10}$ or $\frac{14 \pm \sqrt{196 + 1740}}{10}$) (if completing the square allow as far as shown) or correct values for x or correct values for y dep on correct quadratic
	eg $y = 7 - 2 \times 5$ and $y = 7 - 2 \times 0.6$ (correct labels for x/y)	eg $5.8 = 7 - 2x$ and $-3 = 7 - 2x$ (correct labels for x/y)		M1ft	dep on previous M1 for substituting their 2 found values of x or y in a suitable equation or correct values for the other variable
	<i>Working must be shown</i>		$x = 0.6, y = 5.8$ $x = 5, y = -3$	A1	dep on M1 and the correct quadratic (allow coordinates) must be paired correctly

Answer reference | January 2023 | Question 20

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

				Total 5 marks
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Worked solution

January 2023 | Question 20 | 5 marks

Family: Interpret a system graphically • Variant: Draw the missing line for an intersection

Official final mark-scheme pages are embedded immediately after this question.